

RE-SOLUTION-CONTROLLED FISSION-GAS BEHAVIOR IN UO_2 IRRADIATED IN A FAST FLUX[†]

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Uranium dioxide irradiated in a fast neutron flux to a burnup of 2×10^{20} fissions/cm³ between 650 and 1400°C has been examined by transmission- and scanning-electron microscopy and replication metallography. The fission-gas distribution in the fuel matrix and grain boundaries has been characterized as a function of irradiation temperature and fission rate. The majority of fission gas produced even at the highest irradiation temperature was in the UO_2 matrix either in solution or in the form of bubbles < 20 Å in diameter. The results are explained on the basis of an irradiation-induced re-solution mechanism whereby fission gas from within bubbles is reinjected into metastable solution in the UO_2 lattice. Calculated fission-gas solubilities are given as a function of temperature for 10^{13} , 3×10^{13} , and 10^{14} fissions/cm³ · sec, and, based on these results, it is concluded that the re-solution process is operative over a substantial fuel volume of both light-water-reactor and fast-breeder-reactor oxide fuels.

Le dioxyde d'uranium irradié dans un flux de neutrons rapides à un taux de combustion de 2×10^{20} fissions/cm³ entre 650 et 1400°C a été examiné par transmission et par microscopie électronique par balayage et par métallographie par réplique. La distribution des gaz de fission dans la matrice du combustible et aux joints de grains a été caractérisée en fonction de la température d'irradiation et du taux de fission. La majorité des gaz de fission produits même à la température la plus élevée d'irradiation, se trouvait dans la matrice de UO_2 soit en solution soit sous la forme de bulles de diamètre inférieur à 20 Å. Les résultats sont expliqués sur la base d'un mécanisme de redissolution induite par l'irradiation, mécanisme par lequel le gaz de fission provenant des bulles est réinjecté dans le réseau de UO_2 en solution métastable. Les solubilités calculées des gaz de fission sont données en fonction de la température pour 10^{13} , 3×10^{13} et 10^{14} fissions/cm³ · sec. En se basant sur ces résultats on en conclut que le processus de redissolution est efficace dans un volume important du combustible à la fois pour les combustibles à base d'oxyde utilisés dans des réacteurs à eau légère et dans des réacteurs rapides du type breeder.

Urandioxid wurde in einem schnellen Neutronenfluss bis zu einem Abbrand von 2×10^{20} Spaltungen/cm³ zwischen 650 und 1400°C bestrahlt und transmissions- und rasterelektronenmikroskopisch sowie metallographisch nach dem Abdruckverfahren untersucht. Die Spaltgasverteilung in der Brennstoffmatrix und den Korngrenzen wird in Abhängigkeit von der Bestrahlungstemperatur und der Spaltungsrate angegeben. Der Hauptanteil des gebildeten Spaltgases befindet sich auch bei der höchsten Bestrahlungstemperatur in der UO_2 -Matrix, entweder gelöst oder als Blasen < 20 Å im Durchmesser. Die Ergebnisse werden auf der Basis eines bestrahlungsinduzierten Auflösungsmechanismus erklärt, wobei das in den Blasen befindliche Spaltgas wieder in das UO_2 -Gitter als metastabile Lösung eingebaut wird. Die berechnete Spaltgaslöslichkeit wird in Abhängigkeit von der Temperatur für 10^{13} , 3×10^{13} und 10^{14} Spaltungen/cm³ · s angegeben. Aus diesen Ergebnissen folgt, dass der Auflösungsprozess in Oxidbrennstäben für Leichtwasser- und Schnellbrüterreaktoren in einem grossen Brennstoffbereich wirksam ist.

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1. Introduction

The behavior of inert-gas atoms produced during fissioning of uranium and plutonium in nuclear fuels plays an important role in determining the swelling behavior and fission-gas-release rates from ceramic fuels. Under conditions of thermodynamic equilibrium, xenon and krypton atoms have extremely low solubilities in UO_2 and in other materials [1, 2]. However, recent experiments with irradiated oxide fuel [3–11] have shown that the thermodynamic solubilities of the fission gases obtained from out-of-reactor annealing experiments are not indicative of the amount of gas that can remain in supersaturated solid solution in a fissioning environment.

During irradiation, the passage of energetic fission fragments close to or through a gas bubble imparts sufficient kinetic energy to the gas atoms to knock them into solution in the matrix. This process is irradiation-induced re-solution of the fission-gas bubbles [3, 7–13], and leads to a higher effective fission-gas solubility. Since the size of the inert-gas bubbles depends on the fraction of gas atoms in solution and the gas-bubble coalescence rate is strongly dependent on the bubble size, it is important to establish the extent of re-solution in oxide fuels for various irradiation temperatures and fluxes so that bubble size distributions can be calculated. In addition, swelling and gas-release rates depend on the distribution of gas either in solution or in bubbles, which depends on the extent of re-solution.

The majority of experiments that demonstrate the re-solution effect in oxide fuels [3–11] have been performed in a thermal flux at irradiation temperatures $< 1200^\circ\text{C}$. In this study, the fission-gas distribution in UO_2 , which was irradiated in a fast flux (1.9×10^{15} n/cm² · sec) in EBR-II to 1400°C , was investigated. The fission-gas-bubble size distribution, both in the matrix and in the grain boundaries, has been characterized as a function of irradiation temperature and fission rate. Irradiated specimens also were annealed out-of-reactor, and the resultant fission-gas-bubble distribution was compared with the as-irradiated distribution. These results have been analyzed to determine the efficiency of the re-solution process as a function of temperature and to interpret the behavior of fission gases in oxide fuel.

2. Experimental method

The fission-gas distribution in irradiated UO_2 was determined by using transmission- (TEM) and scanning-electron microscopy (SEM) and replication metallography of fractured fuel specimens. The fuel examined in the present investigation was irradiated in EBR-II for 1900 h, and its irradiation history is given elsewhere [14]. The fuel element chosen for this experiment was unique since its thermal history during irradiation was recorded by a thermocouple in the center of the fuel column and by a second thermocouple attached to the cladding. The data obtained from the thermocouples provided a complete thermal history of the element and an experimental check on the calculated temperature profile across the fuel radius.

The fuel used in the subassembly (44.5% ^{235}U enrichment) was cold pressed and sintered into $\frac{1}{4}$ in (6.35 mm) dia UO_2 pellets. The density of the fuel was 93 to 96.4% of theoretical and had an initial oxygen-to-metal (O/M) ratio of 1.999. The element had a maximum linear heat rating of 266 W/cm and was irradiated to ≈ 1 at % burnup. A burnup analysis performed on a specimen taken at the location of peak fission rate showed the burnup to be 0.84 at %. Based on the longitudinal variation of power along the element, burnup in the bottom section was estimated to be 0.73 at %. Using the THTB heat-transfer code, Cushman [15] calculated the radial temperature distribution in the fuel elements at 22 different elevations.

Since the mean lifetime of a submicron-size fission-gas bubble in the fuel matrix is only a few hours [9, 10], the temperature profile of the fuel element during the last few hours of operation is of primary interest in the present investigation. Therefore, the temperatures referred to here are based on the temperature history during the last 72 h at power. Fig. 1 shows the fuel-element centerline temperature, surface temperature, and the fuel fission rate versus distance from the bottom of the fuel column [16].

Two cross sections, one taken from near the bottom of the fuel column and the other at midlength near the peak-temperature and fission-rate location, were chosen for examination to cover a wide range of temperatures at two fission rates. The location of these sections is indicated in fig. 1. Fig. 2 shows a

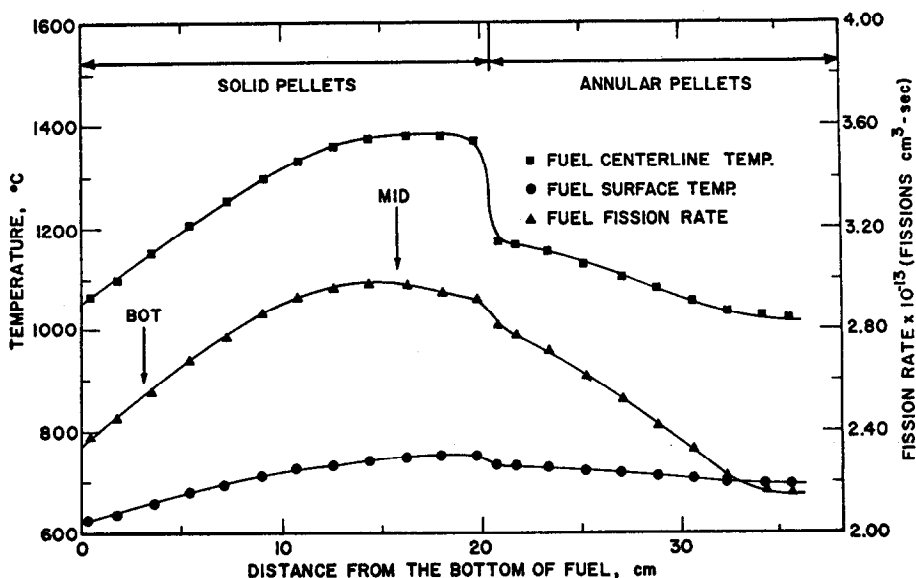


Fig. 1. Calculated UO_2 fuel temperatures and fission rates in fuel element irradiated in EBR-II to a maximum burnup of 0.84 at%. The term BOT indicates the location of the specimens examined from the bottom cross section of the fuel element, and MID indicates the location of those from the peak-temperature and fission-rate cross section.

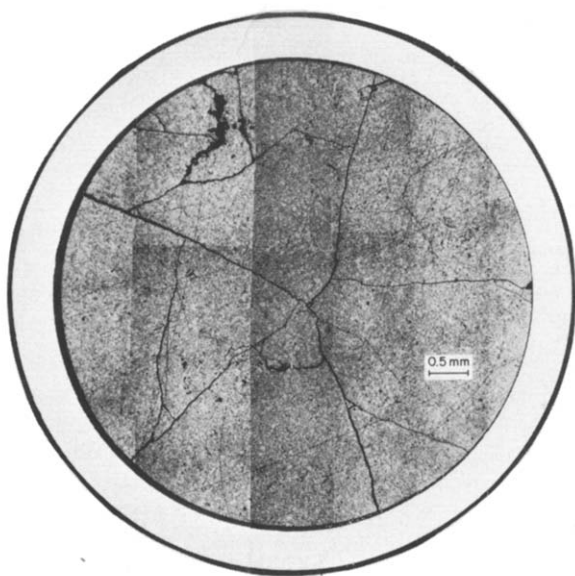


Fig. 2. Composite cross section of UO_2 fuel element close to BOT shown in fig. 1. Cracks, which formed on cooldown, are filled with araldite and appear black. Note the uniform UO_2 grain size over the entire cross section.

typical composite micrograph of a section located 3.2 cm from the bottom of the fuel element (BOT in fig. 1).

A modification of a technique, previously described by Michels et al. [17], was used to prepare the TEM specimens from the as-irradiated fuel. The details of this procedure are given by Pati et al. [18]. By carefully orienting the specimen during the thinning operation, the temperature at a specific location could be determined to within $\pm 50^\circ\text{C}$. In the case of the postirradiation-annealed specimens, a 50 mil dia, 60 mil long cylinder was placed in a tungsten specimen holder and annealed in a Marshall horizontal tube furnace at 1500°C for 5 h in a helium–5% hydrogen atmosphere. The annealed cylinder was chemically thinned, in the same manner as the as-irradiated fuel section.

For chemical polishing, Manley [19] solution Type C (20 ml of orthophosphoric acid, 10 ml of glacial acetic acid, and 2 ml of nitric acid) was used at room temperature. The thinned specimens were transferred to a small glovebox attached to a 200 kV Hitachi HU–200E electron microscope.

Fuel sections, adjacent to the sections that were

thinned for TEM, were fractured and used to prepare replicas for electron microscopy using the technique described by Michels and Dragel [20]. Fractured specimens were also examined by SEM. However, the high beta activity of the specimens (≈ 20 R/h at 5 cm), reduced the resolution attainable with the SEM.

3. Results

Two regions of the fuel element were examined by means of electron and scanning-electron microscopy. One region was near the bottom of the fuel element and the other was from the midlength near the peak temperature and fission-rate zone, as shown in fig. 1. The fission rates in these two regions were 2.6×10^{13} fissions/cm³ · sec and 3×10^{13} fissions/cm³ · sec, respectively. To cover the widest range of temperatures in each region, specimens were taken from three radial locations: rim specimens near the outer diameter of the fuel cross section, middle specimens from a midradial location, and center specimens from the center of the fuel section.

3.1. Fuel specimens from near the bottom of the element: 0.73 at% burnup

3.1.1. Rim specimens irradiated between 650 and 850°C

The range of irradiation temperatures of the rim specimens in the bottom region of the fuel element was from 650 to 850°C. Observation of thin films at this location by TEM showed isolated dislocations and a few grain boundaries. No gas bubbles were observed either in the fuel matrix or on the grain boundaries. However, large pores, 0.5–2.0 μ m in diameter, were seen in scanning-electron micrographs of specimens from the bottom region. These pores, often polyhedral in shape and, for the most part, lying on grain boundaries, are believed to be remnants of the original sintering pores. The same type and size of sintering pores were seen, again chiefly on grain boundaries, in replica electron micrographs.

Fig. 3 is a replica electron micrograph, and all later photomicrographs not identified as transmission- or scanning-electron micrographs are replica electron micrographs. The void volume of pores on the grain boundaries was small, and the fracture was primarily

transgranular. The reduction in pore volume in the as-irradiated condition compared to a starting pore volume of $\approx 6\%$ is believed to be caused by in-reactor densification of the fuel, as has been observed in UO_2 irradiation in several light-water reactors. Higher magnification micrographs from similar transgranular regions revealed only a few pores or fission-gas bubbles less than a few tenths of a micron in size.

3.1.2. Middle specimens irradiated between 850 and 1050°C

The irradiation temperature of the middle specimens in the bottom region of the fuel element was between 850 and 1050°C. Transmission-electron micrographs of fuel from this location showed the as-irradiated dislocation structure and a few gas bubbles ≈ 100 Å in diameter in several localized regions. Since no population of gas bubbles was observed, these few bubbles are believed to be random gas bubbles that nucleated close to sintering pores in the initial material, where a high supersaturation of gas atoms resulted because of re-injection of the sintering gas by the re-solution process. This observation agrees with that of Cornell and Turnbull [10] who noted a considerable variation in local gas-bubble concentration within a sample irradiated to 1.7×10^{19} fissions/cm³. However, it is possible that the random gas bubbles are remnants of sintering pores present in the initial material.

Some of these specimens were annealed for 5–7 sec in the electron beam by removing the condenser aperture of the electron microscope. During beam annealing many fission-gas bubbles precipitated throughout the fuel matrix. Most of the bubbles were uniformly distributed throughout the matrix, were 40–100 Å in diameter, and had a concentration of 1.2×10^{16} bubbles/cm³ (assuming a foil thickness of 1000 Å). Bubbles up to 400 Å in diameter were observed along dislocation lines and on grain boundaries.

A specimen was furnace annealed at 1500°C for 5 h under a helium–5% hydrogen atmosphere to cause precipitation of the fission-gas atoms. From the work of Cornell [21], complete precipitation of the fission-gas atoms into bubbles was expected under these annealing conditions. After annealing, many fission-gas bubbles were seen in the specimen (fig. 4). Uniformly distributed matrix bubbles 40–100 Å in diameter had a concentration of 2.4×10^{16} bubbles/cm³

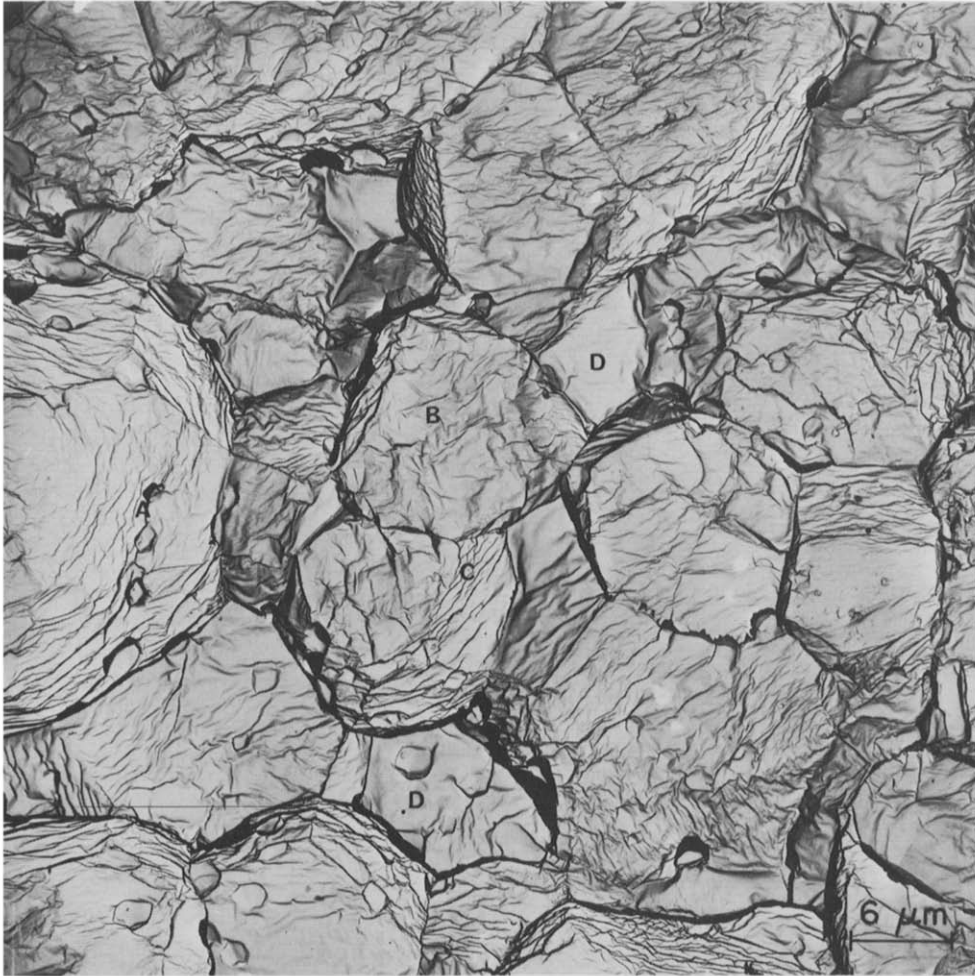


Fig. 3. Micrograph of UO_2 irradiated at $\approx 750^\circ\text{C}$ to 0.73 at% burnup. Typical transgranular fractured region and polyhedral sintering pores on grain boundaries. Polyhedral sintering pores are labeled A, transgranular cleavage areas B, cleavage steps C and intergranular fracture areas D.

(assuming a foil thickness of $1000 \text{ \AA}$). Larger bubbles to $300 \text{ \AA}$ in diameter were observed along dislocation lines.

Scanning-electron micrographs of as-irradiated specimens in this region showed only a few large pores ($0.5\text{--}3.0 \mu\text{m}$ dia, primarily on grain boundaries) that appeared to be more spherical than those irradiated at a lower temperature in the rim location. However, the fracture was almost entirely transgranular and these areas contained almost no pores or gas bubbles. This observation is corroborated by the replica micro-

graphs of fig. 5, which show almost complete transgranular fracture.

3.1.3. Center specimens irradiated above 1050°C

The irradiation temperature of the center specimens in the bottom region of the fuel element was between 1050 and 1150°C . Transmission-electron micrographs from this location showed random dislocation lines and black dots of surface contamination but no fission-gas bubbles $> 25 \text{ \AA}$, which was the limit of resolution in these micrographs. Replica micrographs

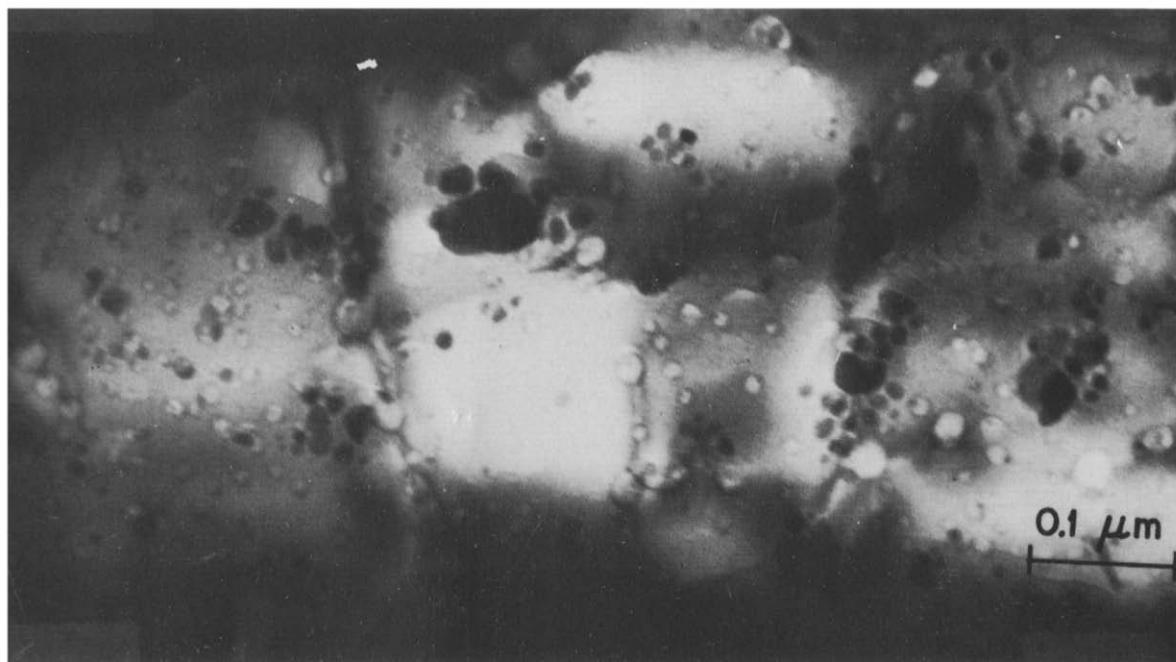


Fig. 4. Transmission-electron micrograph of UO₂ irradiated at $\approx 1050^\circ\text{C}$ to 0.73 at % burnup and furnace annealed at 1500°C for 5 h. Fission-gas bubbles 40 to 100 Å in diameter are distributed throughout the fuel matrix and bubbles to 300 Å in diameter are visible along dislocation lines.

taken from the same location (fig. 6) showed that the fracture was predominantly along grain boundaries, which were decorated with large gas bubbles 0.2–2.0 μm in diameter. The appearance of the bubbles in the scanning-electron micrographs and in the intergranular fracture regions (marked A in fig. 6) suggests that many of the larger grain-boundary bubbles had a lenticular shape. Only a few cases of bubbles that were interconnected to form elongated bubble tubes on the grain boundaries were observed. However, some areas of transgranular fracture that had cleavage steps were seen (B), but no polyhedral sintering pores were observed.

Occasionally some cleavage areas showed the presence of a high concentration of a second population of bubbles ~ 500 Å in diameter. These were believed to be bubbles in the matrix close to the original grain boundaries. They were not true matrix bubbles, since they did not appear uniformly distributed in the matrix and were not observed in the transmission-electron micrographs. As is discussed in subsect. 3.2.3, two ad-

jacent areas were often observed: one area showed these bubbles, whereas the other did not. If these bubbles were uniformly precipitated throughout the matrix, they would appear uniformly distributed on all transgranular fracture areas. Thus, this population of bubbles is believed to originate from gas atoms that were reinjected into the matrix near grain boundaries by re-resolution of gas from the grain-boundary bubbles and sintering pores.

This second population of larger gas bubbles in contrast to the ~ 100 Å dia bubbles observed close to the sintering pores was visible when the cleavage plane ran near and approximately parallel to the grain-boundary plane. Since these bubbles were concentrated near the grain boundaries, which were preferentially dissolved by chemical thinning, these bubbles were polished away and hence were not observed in the transmission-electron micrographs. The bubbles only appeared in replicas when cleavage fracture ran within the grain at a small distance from the boundary.

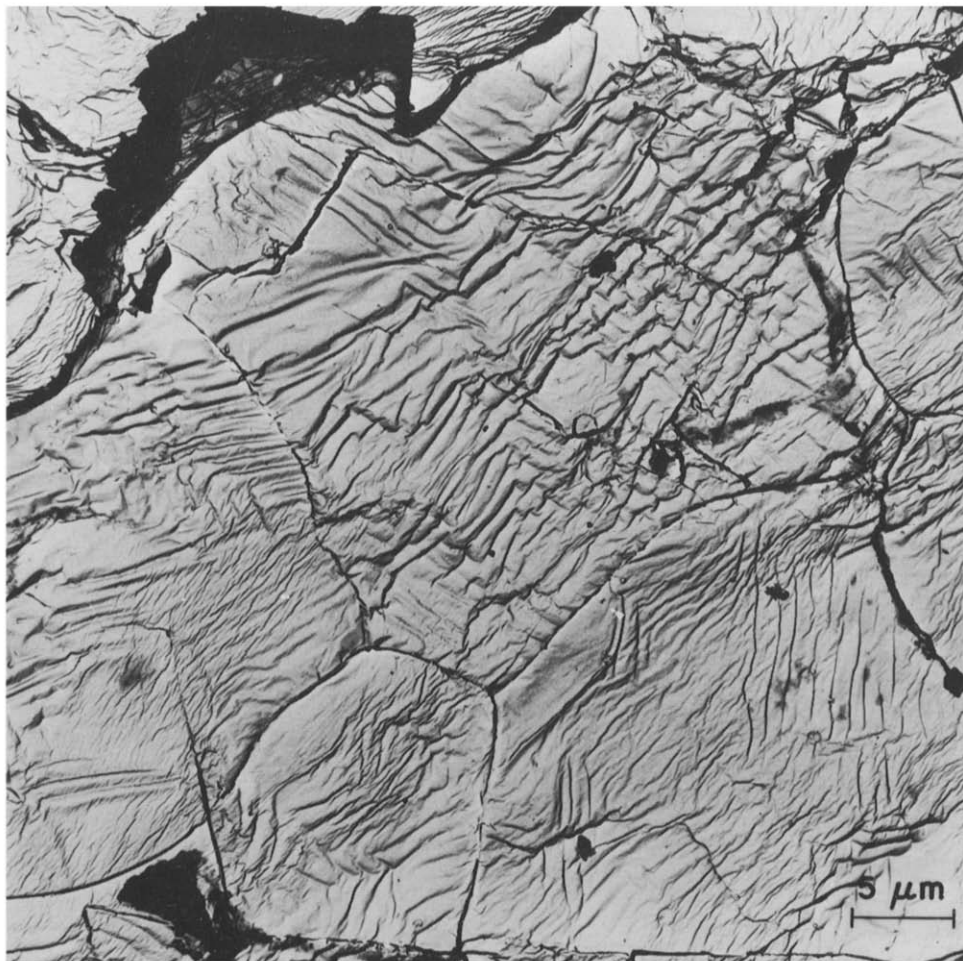


Fig. 5. Micrograph of UO_2 from the same general area as in fig. 4. Structure at a magnification of $4000\times$ shows mostly transgranular fractured surfaces and the absence of pores on the grain boundaries.

3.2. Fuel specimens from the midlength, peak-temperature and fission-rate region of the element: 0.84 at % burnup

3.2.1. Rim specimens irradiated between 700 and 950°C

The irradiation temperature of the rim specimens from the peak-temperature and fission-rate region of the fuel element was between 700 and 950°C . No matrix fission-gas bubbles were observed in transmission-electron micrographs of fuel from this location (fig. 7), above the detection limit of $\sim 20 \text{ \AA}$ in diameter.

Replica micrographs from the same location showed large polyhedral grain-boundary sintering pores $0.5\text{--}8.0 \mu\text{m}$ in diameter, as shown in fig. 8a. Also, a uniformly distributed population of gas bubbles approximately 500 to $1500 \text{ \AA}$ in diameter was present on grain boundaries (A) and on cleavage fractures near grain boundaries (B), as shown in figs. 8a and b. These gas bubbles are believed to have precipitated near the grain boundaries from gas atoms reinjected from the large gas bubbles and sintering pores originally on the grain boundaries as discussed in subsect. 3.1.3. Many of these bubbles were swept by the grain bound-

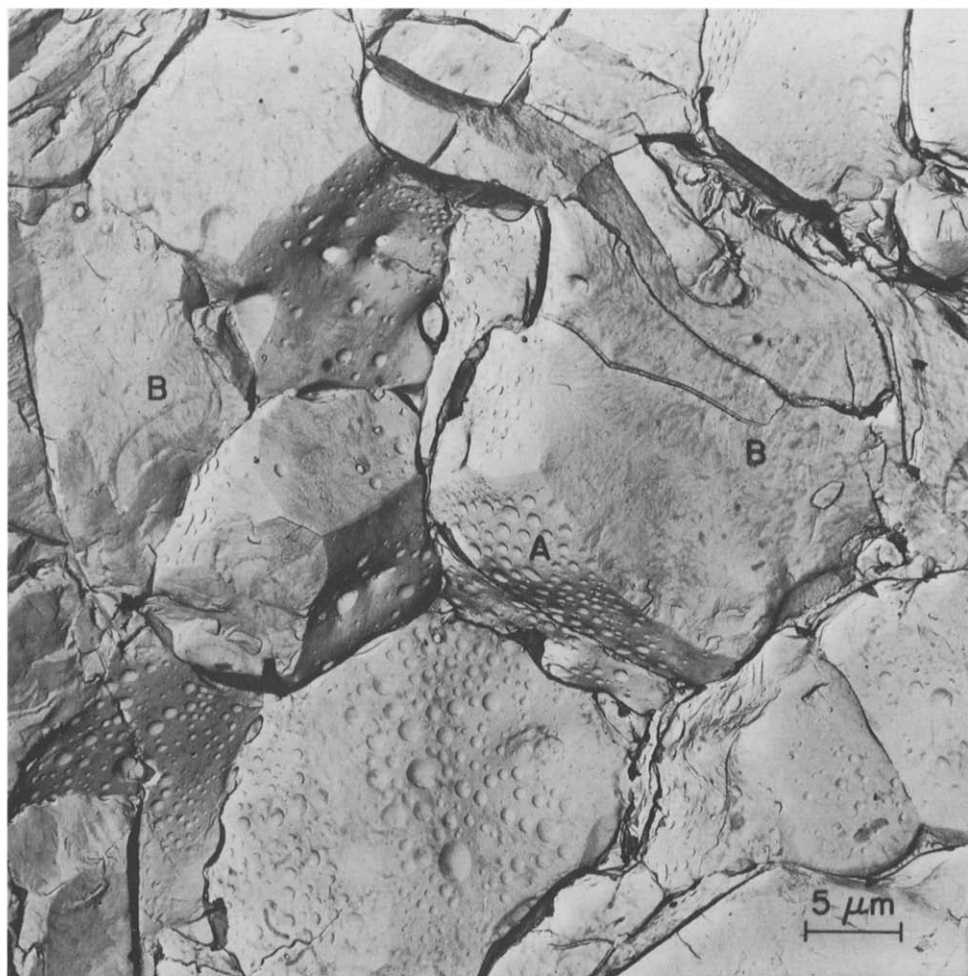


Fig. 6. Micrograph of UO_2 irradiated at $\approx 1100^\circ\text{C}$ to 0.73 at % burnup showing predominantly intergranular fracture areas (A) with a high density of large bubbles on grain boundaries and a few polyhedral voids on boundaries. Note the difference between the structure shown in this figure ($\approx 1100^\circ\text{C}$) and in fig. 3, in which only polyhedral sintering pores were visible at the lower irradiation temperature ($\approx 750^\circ\text{C}$). Some regions of transgranular fracture (B) with cleavage steps are also visible.

aries as the grains grew and are therefore seen on the grain boundaries [22–24].

The difference in the fission-gas distribution between the specimens in this fuel region and those from the bottom region, may be explained in part by the $\approx 20\%$ higher burnup in this region resulting in a 20% higher concentration of fission-gas atoms. Since the fission rate in this region was $\approx 20\%$ higher than at the bottom of the fuel element, more gas atoms were reinjected from grain boundaries into the nearby

matrix. Also, a small amount of grain growth was observed in the middle section compared to almost no grain growth in the bottom section. These three factors resulted in a higher concentration of gas bubbles on the grain boundaries than that observed in the bottom section.

3.2.2. Middle specimens irradiated between 950 and 1300°C

The irradiation temperature of the middle speci-

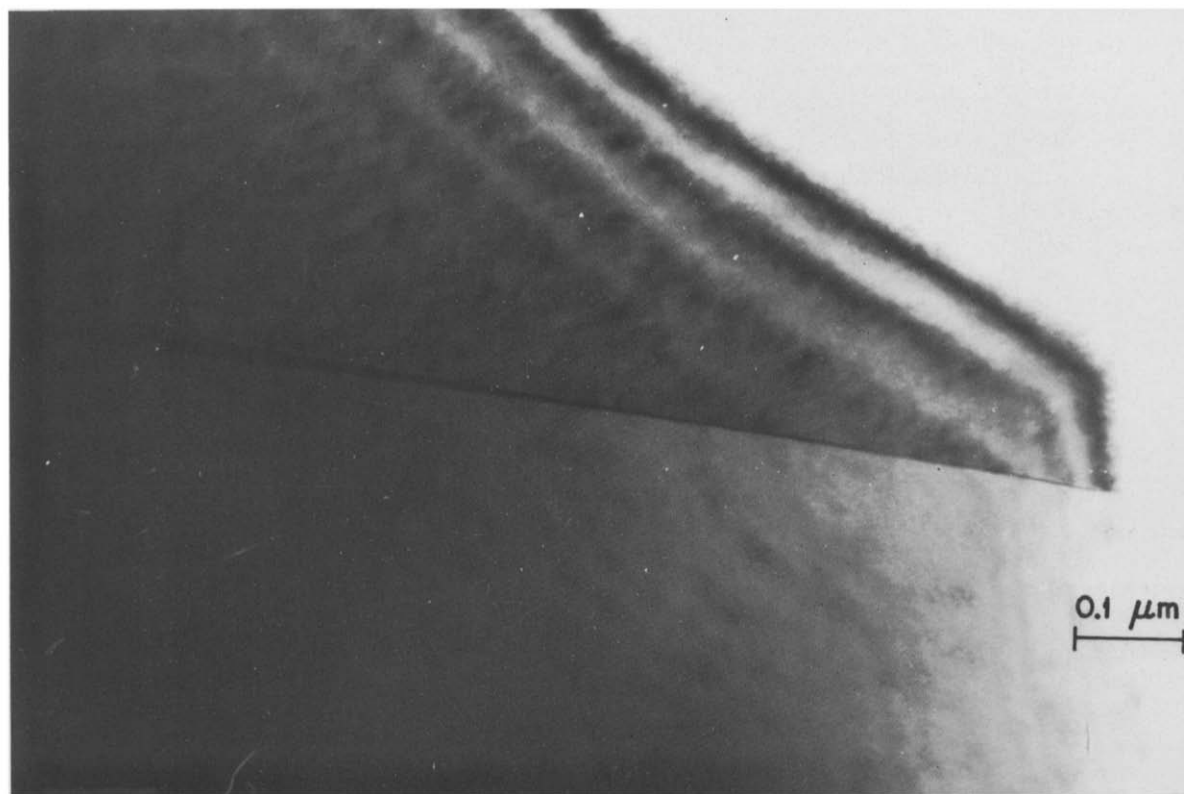


Fig. 7. Transmission-electron micrograph of UO_2 irradiated at $\approx 950^\circ\text{C}$ to 0.84 at % burnup. A grain boundary is visible perpendicular to the edge of the foil, but no fission-gas bubbles $> 20 \text{ \AA}$ in diameter could be detected.

mens from the peak-temperature and fission-rate region of the fuel element was between 950 and 1300°C . A few gas bubbles up to $100 \text{ \AA}$ in diameter were observed near grain boundaries or the etched remnants of pores, but no well-defined population of gas bubbles was observed within the matrix. Some polyhedral sintering pores in replica electron micrographs of the same location were observed. The fracture was predominantly intergranular, and almost all grain boundaries were heavily decorated with gas bubbles. Many large grain-boundary bubbles were observed, some of which were interconnected, as shown in fig. 9. Again, a uniformly distributed population of gas bubbles (500 – $2000 \text{ \AA}$ in diameter) was seen on grain boundaries and cleavage areas near grain boundaries. The bubble size and concentration of this population were different for different cleavage areas, which suggests that the size distribution depends on the dis-

tance between the cleavage and the grain boundary planes.

3.2.3. Center specimens irradiated between 1300 and 1400°C

The irradiation temperature of the center specimens from the peak-temperature and fission-rate region of the fuel element was between 1300 and 1400°C . Most of the transmission-electron micrographs showed a high density of dislocations, but no well-defined population of fission-gas bubbles of $20 \text{ \AA}$ or greater was observed even at the highest irradiation temperature. Replica electron micrographs from this location showed large gas bubbles on grain boundaries, many of which were interconnected forming gas tunnels and pipes (fig. 10). A uniformly distributed population of gas bubbles 500 to $2000 \text{ \AA}$ in diameter was seen on grain boundaries and cleavage areas near

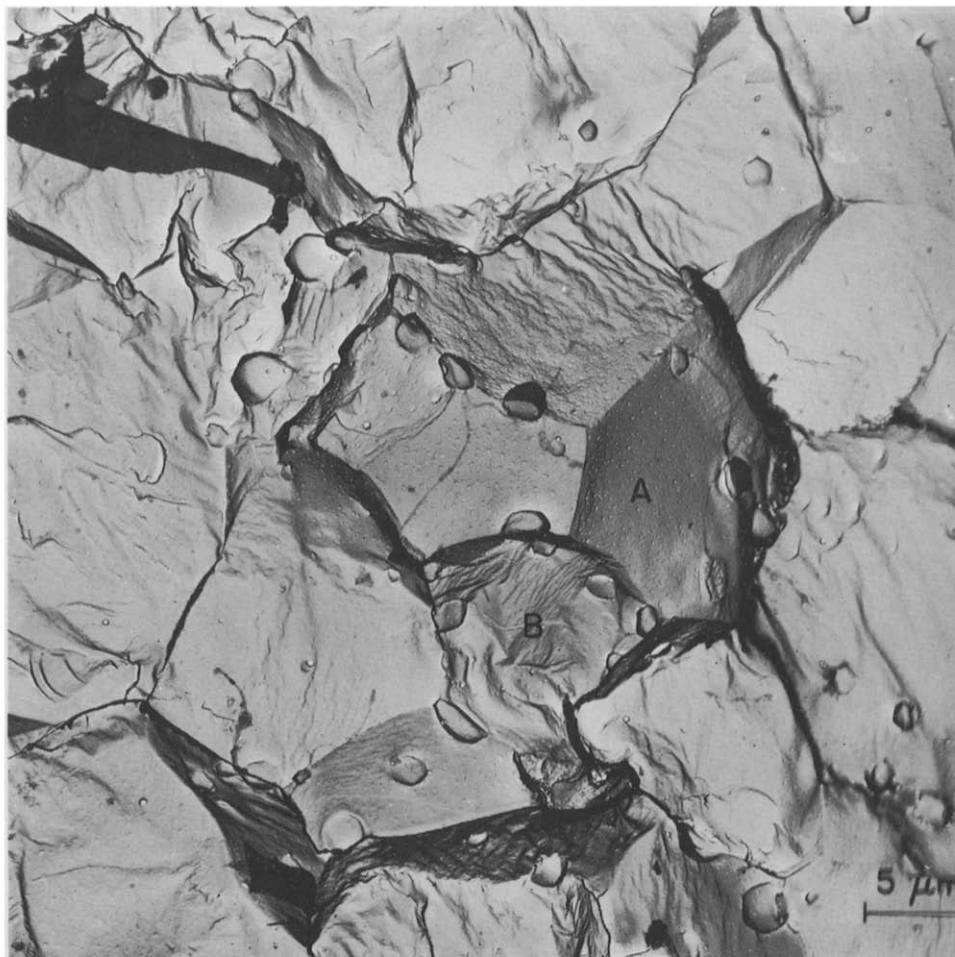


Fig. 8a.

grain boundaries. Again the size and distribution of this bubble population appeared to be dependent on the distance of the cleavage surface from the grain boundary. In many cases, one cleavage area had a high concentration of uniformly distributed bubbles, whereas the cleavage surface of the adjacent grain had only a few gas bubbles. Since the matrix-gas-bubble concentration of adjacent grains should be the same because their irradiation conditions were the same, these observations suggest that the cleavage areas with a high concentration of bubbles are located near grain boundaries.

4. Discussion

4.1. Fission-gas inventory and bubble-size distribution in grains and at grain boundaries

In the peak fission and temperature region, the element had a burnup of 0.84 at %, which is equivalent to $\approx 2 \times 10^{20}$ fissions/ cm^3 of UO_2 . Thus, 5.6×10^{19} xenon and krypton atoms/ cm^3 of UO_2 were produced in the fuel. (In this case, the total yield of xenon and krypton is taken to be 0.28 atoms/atom of ^{235}U fissioned.) From the isotopic analysis of the plenum



Fig. 8b.

Fig. 8. Micrograph of irradiated UO₂ from the same region as in fig. 7. (a) Shows large polyhedral sintering pores on the grain boundaries and uniformly distributed fission-gas bubbles 500 to 1500 Å in diameter on the grain boundaries (A) and on cleavage fractures near the grain boundaries (B). (b) Higher magnification micrograph of a region close to (a) showing the grain boundaries, sintering pores, and a distribution of bubbles 500 to 1500 Å in diameter on and near grain boundaries.

gases following irradiation, 2% of the xenon and krypton produced were found to be released into the plenum, which left 98% of the fission gases retained within the fuel, either in the fuel matrix or in the grain boundaries.

The amount of fission gases accumulated in the grain boundaries was estimated by characterizing the size of the intergranular bubble population. Table 1

lists the parameters evaluated from analysis of micrographs of the intergranular regions. This analysis was limited to sections in which the grain-boundary bubbles were essentially spherical, because the estimation of gas-atom density in bubbles is difficult for nonspherical shapes.

In estimating the number of gas atoms in a bubble, equilibrium of surface tension with internal pressure

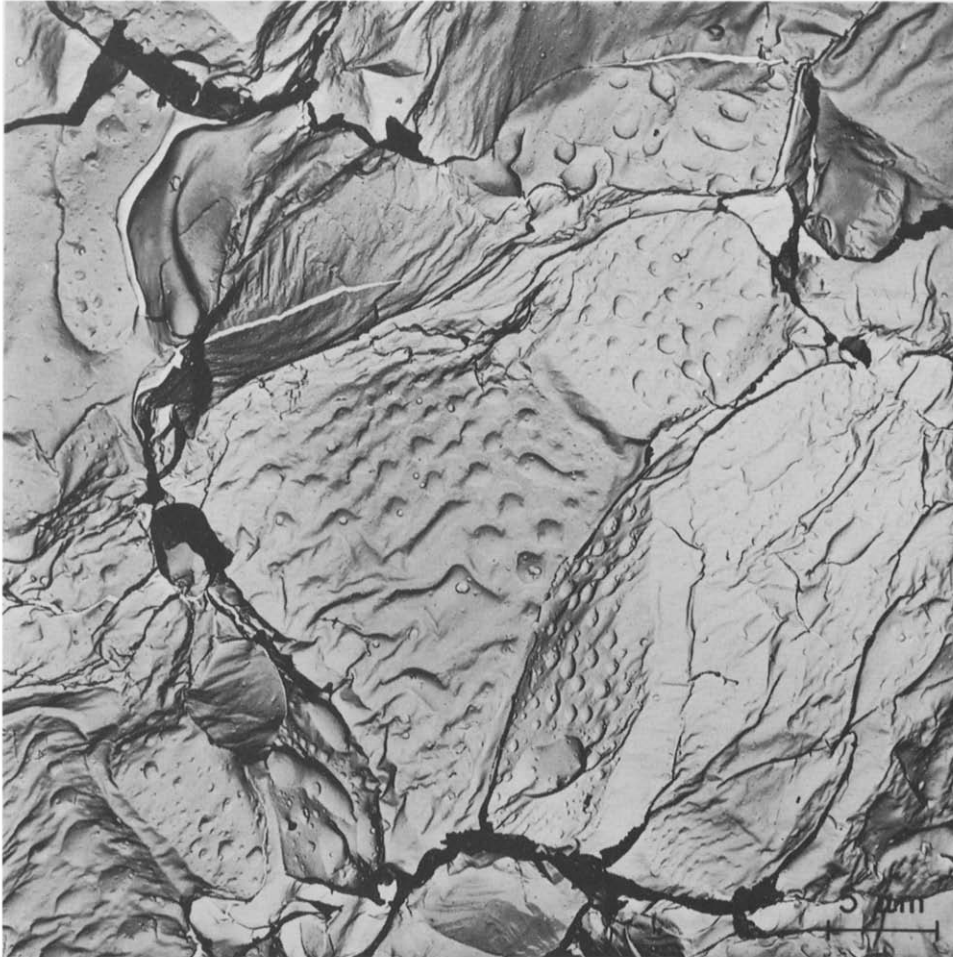


Fig. 9. Micrograph of UO_2 irradiated at $\approx 1150^\circ\text{C}$ to 0.84 at% burnup. Interconnected, large, grain-boundary bubbles and a uniformly distributed population of fission-gas bubbles 500 to 2000 Å in diameter are visible on grain boundaries and cleaved surfaces near the grain boundaries.

was assumed, and Harrison's equation of state for xenon [25] was used. When the bubble diameter is $\geq 0.2 \mu\text{m}$, the difference in the estimated number of gas atoms per bubble, based on the perfect gas law and Harrison's equation of state for xenon, is negligible; hence $m \propto d^2$, where m is the number of gas atoms present in a bubble of diameter d . Since the bubbles exist in a range of sizes, a more accurate estimate of the average number of gas atoms $\bar{m}$ is obtained when the root-mean-square (rms) diameter $(\bar{d}^2)^{1/2}$ is used instead of the mean diameter $\bar{d}$. The rms diameter is

given by

$$(\bar{d}^2)^{1/2} = \left(\frac{\sum N_i d_i^2}{\sum N_i} \right)^{1/2}. \quad (1)$$

The average temperature of each specimen and a surface energy of 600 erg/cm^2 for UO_2 were used [26] in the calculation of $\bar{m}$. The average number of gas atoms per bubble [27] calculated for the measured bubble-size distributions is listed in table 1. Multiplication of $\bar{m}$ by the number of gas bubbles per unit area of grain boundary gives the number of gas

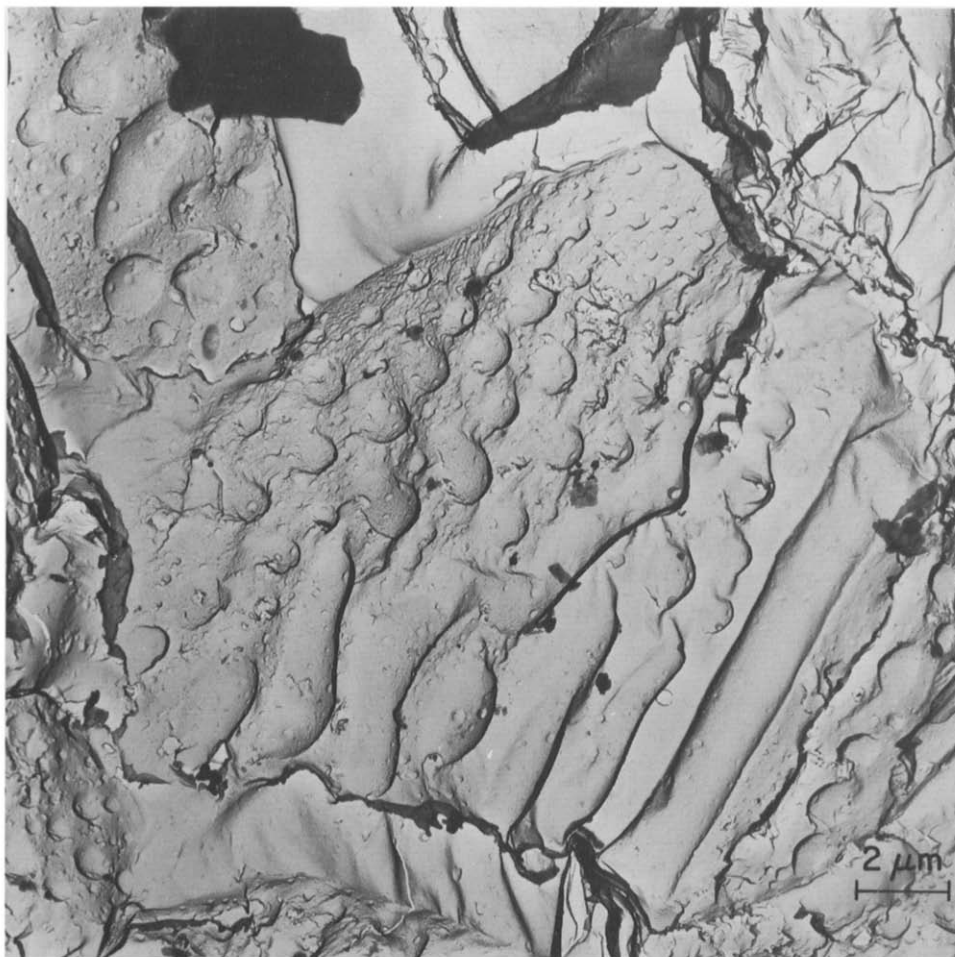


Fig. 10. Micrograph of UO_2 irradiated at $\approx 1350^\circ\text{C}$ to 0.84 at % burnup. Shows extensive interconnection of fission-gas bubbles that form tunnels and pipes along the grain boundaries.

atoms per unit area of grain boundary. To obtain the number of gas atoms accumulated on the grain boundaries per unit volume of fuel, the number of gas atoms accumulated per unit area of grain boundary is multiplied by the grain-boundary area per unit volume S_v . S_v was estimated from measurements of the linear grain-boundary intercept N_L and is given by

$$S_v = 2N_L. \quad (2)$$

The total gas content in grain-boundary bubbles is listed in table 1, as well as an estimate of the total amount of fission gas produced. From table 1, it is evident that only a small fraction of the fission gases

are precipitated in intergranular bubbles. Even up to 1350°C , almost 80% of the fission gas remains in the matrix either in solution or in the form of fission-gas bubbles $< 20 \text{ \AA}$ in diameter.

Other evidence that supports this conclusion is the precipitation of a high density ($2.4 \times 10^{16} \text{ cm}^{-3}$) of intragranular fission-gas bubbles 40 to $100 \text{ \AA}$ in diameter during postirradiation annealing of an irradiated specimen, which did not contain any observable fission-gas bubbles before annealing. Approximately 100% of the fission gases produced could be accounted for in the intragranular gas bubbles precipitated during the postirradiation anneal. Direct evidence for

Table 1
Size distribution and gas content of intergranular fission-gas bubbles in as-irradiated UO_2 .

Parameters	Bottom region of fuel element	Peak-temperature and fission- rate region	
	Center specimen	Rim specimen	Middle specimen
Irradiation temperature ($^{\circ}C$)	1150	750	1350
Fission rate ($\times 10^{13} \text{ cm}^{-3} \cdot \text{sec}^{-1}$)	2.6	3.0	3.0
Range of bubble sizes observed (μm)	0.111–2.19	0.056–0.444	0.3–3.7
Mean diameter (μm)	0.491	0.187	0.733
Root-mean-square diameter (μm)	0.646	0.221	0.998
Average number of gas atoms/bubble ($\times 10^7$ atoms)	2.65	0.42	5.61
Bubble density in grain boundary ($\times 10^8 \text{ cm}^{-2}$)	1.12	7.50	2.07
Grain-boundary area per unit volume (cm^2/cm^3)	1350	1150	1000
Total gas content in grain-boundary bubbles ($\times 10^{18} \text{ atoms/cm}^3$)	7.42	3.62	11.6
Total gas produced ($\times 10^{18} \text{ atoms/cm}^3$)	48	56	56
Gas in grain boundaries (% of total)	15	6	21

the presence of a high degree of supersaturation of fission gases in metastable solution is the formation of fission-gas bubbles *in situ* during electron-beam annealing of an irradiated specimen, which did not contain any observable fission-gas bubbles in the as-irradiated state.

The presence of such a large amount of fission gas in the matrix is the result of the kinetic re-solution effect [3–11, 19, 28–30]. Under thermodynamic equilibrium, xenon and krypton are essentially insoluble in UO_2 [1, 2]. Thus, the formation of fission-gas bubbles by precipitation of gas atoms is expected when atomic mobilities are sufficiently high to allow migration of the gas atoms during the time scale of irradiation. Continued precipitation into gas bubbles and migration and coalescence of these bubbles will lead to growth of the bubbles to sizes that are easily observed by TEM. The presence of most of the fission gas atoms in solution suggests that an efficient re-solution process occurs during irradiation at temperatures to $1300^{\circ}C$. This re-solution process retards the rate of precipitation of gas atoms into bubbles by con-

tinuously reinjecting gas atoms back into the matrix. This results in a supersaturated metastable solution of fission gases in the fuel matrix and in the formation of smaller bubbles than predicted for purely diffusional and collisional growth processes [31].

4.2. Re-solution mechanisms

Several mechanisms have been proposed to account for the irradiation-induced re-solution phenomenon, all of which rely on the passage of energetic particles (high-energy neutrons [12], fission fragments [3, 12], or fission-fragment-induced collision cascades [12]) close to or through a bubble, which imparts sufficient energy to the gas atoms to knock them into solution in the matrix. Based on their recent observations, Turnbull and Cornell [11] proposed that small ($\leq 25 \text{ \AA}$ in diameter) intragranular bubbles may be destroyed in a single encounter with a fission fragment.

Irrespective of the mechanism involved, a re-solution parameter b , defined as the probability per

second that a gas atom undergoes re-resolution [7], describes the rate of re-resolution. If m is the number of gas atoms in a bubble being irradiated for time t , then [7]

$$b = -\frac{1}{m} \frac{dm}{dt}. \quad (3)$$

Turnbull and Cornell [7, 8] have experimentally evaluated b by characterizing the bubble distribution of a sample after postirradiation annealing, re-irradiating (to a dose much smaller than the original dose), and then estimating the change in m during re-irradiation. Measured values of b lie in the range of 10^{-4} to 10^{-5} sec^{-1} for a dose rate of 10^{13} fissions/ $\text{cm}^3 \cdot \text{sec}$ and irradiation temperatures of 200 – 1200°C . Problems in accurately determining the sizes of small bubbles and errors in estimating m make it difficult to reduce the uncertainty in b . However, from their most recent results [11], these authors estimate b to be between $\approx 2 \times 10^{-4}$ and $\approx 4 \times 10^{-4} \text{ sec}^{-1}$, for a dose rate of 10^{13} fissions/ $\text{cm}^3 \cdot \text{sec}$ and an irradiation temperature of 1200°C . Comparing the measured values of b with the calculated values for the various re-resolution mechanisms, Turnbull [13] and Turnbull and Cornell [11] suggested that electron excitation, which results from the passage of a fission track close to a bubble, is the most probable rate-controlling re-resolution process, although the present theories are inadequate to explain the details of the process.

In the present investigation no fission-gas bubbles could be detected in UO_2 irradiated up to 1400°C to 2×10^{20} fissions/ cm^3 . Cornell [9] observed a high concentration (1×10^{17} to $\approx 4 \times 10^{17}$ bubbles/ cm^3) of small (16–31 Å in diameter) fission-gas bubbles in UO_2 irradiated between 3.5×10^{19} and 4.6×10^{20} fissions/ cm^3 at temperatures of 560 to $\approx 1300^\circ\text{C}$. However, even these high concentrations of intragranular bubbles contained only ≈ 15 – 30% of the total fission gas generated. In considering the passage of a fission fragment close to a bubble as the rate-controlling process for re-resolution, b should be proportional to fission rate. Thus a factor of three higher fission rate in our irradiation experiment compared with that of Cornell [9] may have caused the observed difference in the intragranular gas-bubble distributions by increasing the re-resolution rate so that either gas-bubble formation is prevented or the growth of the bubbles is limited to < 20 Å in diameter.

Since the volume contribution to swelling per fission-gas atom is constant from a single gas atom in solid solution to ≈ 1000 gas atoms precipitated in bubbles up to 50 Å in diameter, it is immaterial, from the swelling viewpoint, whether fission gases are present in solid solution or in small bubbles of up to 50 Å in diameter. Consequently, for swelling considerations, individual fission-gas atoms or bubbles < 20 Å in diameter can be considered to remain in solution.

4.3. Irradiation-induced metastable solubility of fission gases

To evaluate the effect of re-resolution on fuel swelling, the kinetic solubility of fission gases in UO_2 has been calculated as a function of fission rate, irradiation dose, and temperature. Based on the formulation of Nelson [12] and inclusion of the loss of fission-gas atoms to grain boundaries, the instantaneous rate of change of the gas concentration per unit volume in the lattice, where c is the gas concentration at irradiation time t , can be described by

$$\begin{aligned} \frac{dc}{dt} &= \left[\begin{array}{c} \text{Rate of change of} \\ \text{gas concentration} \\ \text{in the lattice} \end{array} \right] \\ &= \left[\begin{array}{c} \text{Rate of generation} \\ \text{of gas atoms} \end{array} \right] - \left[\begin{array}{c} \text{Rate of precipita-} \\ \text{tion in bubbles} \end{array} \right] \\ &\quad + \left[\begin{array}{c} \text{Rate of re-resolution} \\ \text{of gases from} \\ \text{bubbles} \end{array} \right] - \left[\begin{array}{c} \text{Rate of loss of} \\ \text{gas atoms to grain} \\ \text{boundaries} \end{array} \right]. \end{aligned} \quad (4)$$

As shown above only a small fraction of the fission gases are contained in bubbles on the grain boundaries up to an irradiation temperature of 1350°C . Ronchi and Matzke [32] calculate that the fraction of fission gases precipitated at grain boundaries is only a small fraction (0.18) of the gas generated to an irradiation temperature of 1200°C and a dose of 7×10^{20} fissions/ cm^3 . The fraction of gas precipitated at grain boundaries increases sharply with temperature and is ≈ 0.9 at 1500°C for the same dose. However, the calculations of Ronchi and Matzke are based on the Booth model [33] and the results are expected to be high since the re-resolution of gas atoms from grain-boundary bubbles is not considered [34]. In the present case, for a maximum fuel temperature of

$\approx 1400^\circ\text{C}$, the fourth term in eq. (4) is neglected in describing the behavior of the fission gas in the intragranular region.

The instantaneous gas-generation rate is given by $F\dot{f}$, where F is the number of gas atoms produced per fission (taken equal to 0.28), and $\dot{f}$ is the fission rate. From Ham [35], the gas-atom precipitation rate in spherical bubbles of radius R spaced $2a$ apart is given by $3RD_g c/a^3$, where D_g is the gas-atom diffusion coefficient. The re-solution rate per unit volume is given by $(bm)n$, where n is the bubble density per unit volume, and m is the number of gas atoms in a bubble, as defined above.

Substituting the above in eq. (4) yields

$$\frac{dc}{dt} = F\dot{f} - \frac{3RD_g c}{a^3} + (bm)n. \quad (5)$$

Since the release of gas from UO_2 is negligible and the amount of gas precipitated in the grain boundaries is relatively small to 1300°C , based on the conservation of gas atoms in the fuel, we have

$$F\dot{f}t = c + nm. \quad (6)$$

Substituting $n = 3/4\pi a^3$ and noting that, for small bubbles, $m \approx 4\pi r^3/3v$, where v is the van der Waals' constant for xenon ($8.5 \times 10^{-23} \text{ cm}^3$), and rearranging the terms gives

$$\frac{dc}{dt} = F\dot{f} + b(F\dot{f}t - c) - (4\pi n)^{2/3}(3v)^{1/3}D_g c(F\dot{f}t - c)^{1/3}. \quad (7)$$

Eq. (7) can be numerically integrated to obtain c as a function of irradiation dose. However, to obtain a closed-form solution of c , a quasi-steady-state solution is obtained by setting $dc/dt = 0$ for short time intervals at various irradiation times. This solution assumes that, during the short time interval at irradiation time t , the rate of capture of the fission gases by the bubbles is equal to the rate of generation plus the rate of re-solution. The concentration of gas atoms is then given by [36]

$$c = F\dot{f}t - \frac{1}{b} [(4\pi n)^{2/3}(3v)^{1/3}D_g c(F\dot{f}t - c)^{1/3} - F\dot{f}]. \quad (8)$$

Using eq. (8), the concentration of fission gas in solution has been calculated as a function of irradiation dose and temperature for a fission rate of 3×10^{13} , as shown in fig. 11. The values of D_g and n measured by Cornell [9] and Turnbull and Cornell [7, 8, 10] for

a fission rate of $\approx 10^{13}$ have been used as base values. Below 1300°C , D_g was varied linearly with fission rate since irradiation-enhanced diffusion has been observed by Cornell [9] to this temperature. Above 1300°C , thermal-diffusion-controlled values of D_g as measured by Cornell [21] have been used. The bubble density per unit volume n has been kept constant at 10^{17} because experimentally n has been found to be insensitive to the irradiation conditions. The re-solution parameter b has been taken proportional to fission rate, since the fission-fragment-gas-bubble collision is the most efficient re-solution process [11, 13]. The calculated concentrations of fission gas in solution for values of $b = 10^{-4}$ and 10^{-5} sec^{-1} under 3×10^{13} fissions/ $\text{cm}^3 \cdot \text{sec}$ are shown in fig. 11, since the range of b values measured by Turnbull and Cornell [7, 8, 11] was within these limits.

The solid line in the graph represents the total

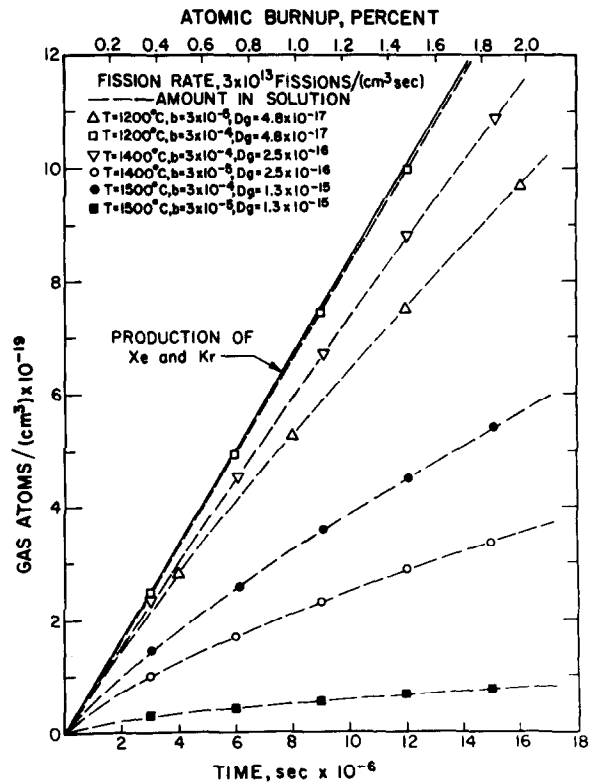


Fig. 11. Calculated amount of fission gas in solution in UO_2 as a function of irradiation dose and temperature for a fission rate of 3×10^{13} fissions/ $\text{cm}^3 \cdot \text{sec}$.

amount of fission gas produced, and the dashed lines represent calculated amounts of gas in solution for various diffusion coefficients and re-solution parameters. As seen in fig. 11, to 1200°C and ≈ 1 at % burnup, almost all fission gas remains in solution for values of b from 3×10^{-5} to $3 \times 10^{-4} \text{ sec}^{-1}$.

However, the calculated amount of gas in solution at 1400°C drops to $\approx 30\%$ of that produced for $b = 3 \times 10^{-5} \text{ sec}^{-1}$. For $b = 3 \times 10^{-4} \text{ sec}^{-1}$, the amount of gas in solution at 1400°C remains at 90% of the amount produced. Since to 1400°C we do not see measurable fission-gas bubbles, our experimental results are consistent with a re-solution parameter equal to $3 \times 10^{-4} \text{ sec}^{-1}$. For $b = 3 \times 10^{-4} \text{ sec}^{-1}$ at 1500°C , the amount of gas in solution is $\approx 50\%$ of the amount produced. However, in an evaluation of the calculated results above 1400°C , it must be noted that grain-boundary loss has been neglected, and the loss to grain boundaries becomes substantial above 1400°C . Thus eq. (8) is no longer valid.

It is to be noted that, in the above calculation of fission gases in solution, it was assumed there are 10^{17} fission-gas bubbles/ cm^3 of fuel which act as sinks for

precipitation of gas atoms. It is interesting to calculate the number of fission-gas atoms each of these intragranular bubbles contain at the highest irradiation temperature of 1400°C . At a dose of 2×10^{20} fissions/ cm^3 , the amount of fission gases in solution at 1400°C is calculated to be 5.1×10^{19} (fig. 11) of a total of 5.6×10^{19} gas atoms generated per cubic centimeter of fuel, which leaves only 5×10^{18} ($\approx 10\%$ of total) gas atoms to be precipitated as intragranular bubbles. Taking into account $\approx 25\%$ loss of gas atoms to grain boundaries (table 1, at 1350°C), 3.75×10^{18} gas atoms are expected to be present in 10^{17} intragranular bubbles. Thus, on an average, each gas bubble contains ≈ 37 atoms. At 1400°C , the equilibrium size of a bubble that contains 37 gas atoms is estimated to be $17 \text{ \AA}$ in diameter, which is below the detection limit of the present investigation. Thus, this estimate agrees with the observation that the 10^{17} intragranular bubbles assumed to be present in the fuel were smaller than the size detectable in TEM examination and therefore were not observed.

A detailed calculation [31] based on the modified forms of the GRASS code [37] was performed to

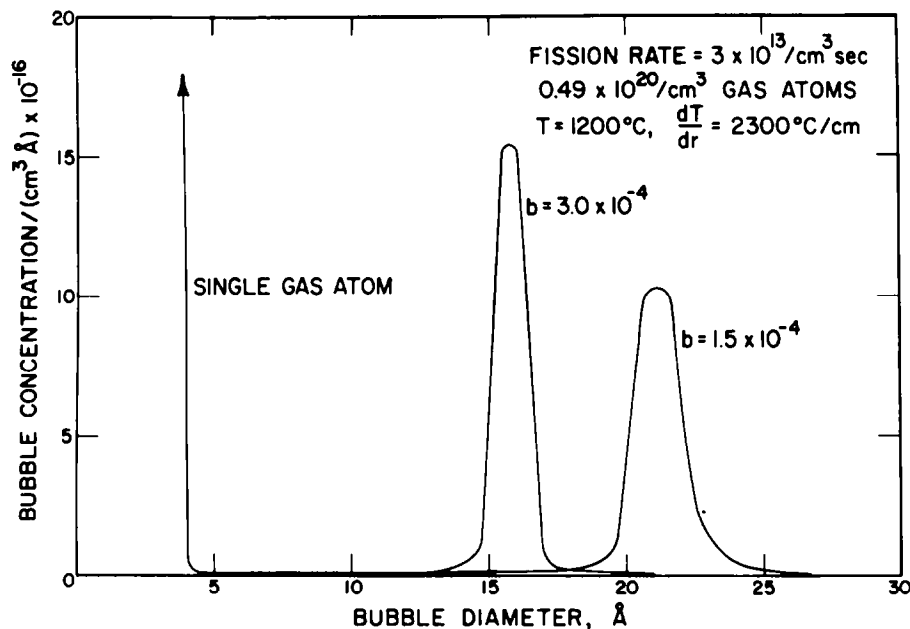


Fig. 12. Calculated bubble-size distribution for two values of the re-solution parameter b for a fission rate of 3×10^{13} fissions/ $\text{cm}^3 \cdot \text{sec}$ at 1200°C .

determine the evolution of the intragranular bubble-size distribution, taking into account bubble coalescence and gas atom re-solution. As shown in fig. 12, for the irradiation conditions of this investigation, a peak in the distribution curve of bubble number density in a unit bubble-size range versus bubble diameter is observed at a bubble diameter of $\approx 16 \text{ \AA}$. The number density is found to go down drastically on either side of the peak size. Thus, the result of this calculation is also consistent with the observation that the intragranular bubbles, if present, were below the detectability limit of the experimental technique used in the present investigation.

Figs. 13a and 13b show the calculated amounts of gas in solution at various irradiation temperatures, using the D_g and b values listed in the figures, for fission rates of 10^{13} and 10^{14} fissions/ $\text{cm}^3 \cdot \text{sec}$, respectively. At 10^{13} fissions/ $\text{cm}^3 \cdot \text{sec}$, which corresponds to the fission rate in Cornell's experiments [9],

the calculated values agree reasonably well with the values measured experimentally (i.e., the amount of gas produced minus the amount in bubbles larger than $17 \text{ \AA}$ in diameter). In general, the measured amount of gas in solution quoted by Cornell [9] covers a range of irradiation temperatures. However, of particular interest is the value quoted for 1065°C , and, as is evident, the agreement between our calculated value (100% in solution) and the value measured experimentally (90% in solution) is good.

Calculations are shown in fig. 13b for 10^{14} fissions/ $\text{cm}^3 \cdot \text{sec}$, since fuel elements for the FFTF are designed to have a peak fission rate in this range. At this fission rate and 1400°C , $\approx 94\%$ of the fission gas is calculated to remain in solution up to 10 at % burnup (for an assumed re-solution parameter of 10^{-3} sec^{-1}). At 1600°C , the calculated amount of gas in solution decreases to about 25% of the amount produced at 10 at % burnup. The amount of gas in

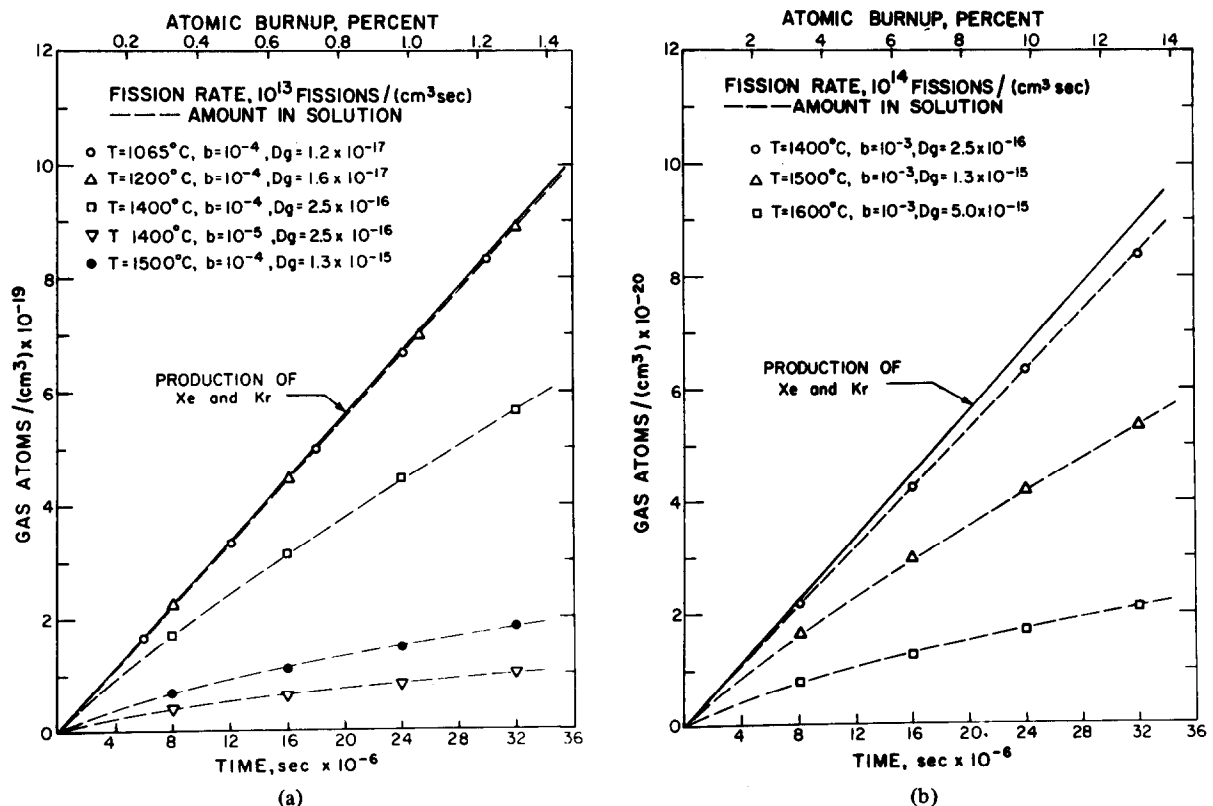


Fig. 13. Calculated amount of fission gas in solution in UO_2 as a function of irradiation dose and temperature for a fission rate of (a) 10^{13} fissions/ $\text{cm}^3 \cdot \text{sec}$, and (b) 10^{14} fissions/ $\text{cm}^3 \cdot \text{sec}$.

solution will be lower than 25% since the loss of gas atoms to grain boundaries was neglected in the above estimate.

Thus, the temperature dependence of re-solution can be predicted for any fission rate and burnup of interest. In addition, from the analysis of the measured gas distribution in the present experiment, it is possible to establish a value of b that is both consistent with our measurements and with other published results. This value of b is 10^{-4} at 10^{13} fissions/cm³ · sec, which is close to the value suggested by Turnbull and Cornell [11].

A value of b of 10^{-4} sec⁻¹ means that, under a fission rate of 10^{13} fissions/cm³ · sec, bubble lifetimes are only a few hours. Thus, the fuel temperature and fission rate during the last few hours of irradiation control the fission-gas distribution that is observed during postirradiation examination. Although no measurement of b has been performed in mixed-oxide fuel, it is reasonable to expect b to be in the same range for both UO₂ and mixed-oxide fuel. Since a large volume of fuel is irradiated below 1400°C under typical LMFBR operating conditions and no fission-gas bubbles are observed up to this temperature in UO₂, we conclude that the re-solution process will play a major role in controlling the fission-gas behavior in LMFBR and LWR fuel elements.

5. Conclusions

The majority of the fission gases produced in uranium dioxide irradiated in a fast neutron flux to 2×10^{20} fissions/cm³ between 650 and 1400°C have been found to be in the matrix, either in metastable solution or in the form of bubbles < 20 Å in diameter. Only 6 to 21% of the fission gases were found in bubbles and pores on the grain boundaries between 700 and 1350°C. These results are explained on the basis of irradiation-induced re-solution of fission gases, which reinjects gas from within the gas bubbles to solution in the lattice and hence retards the growth rate of bubbles.

The irradiation-induced kinetic solubility of fission gases in UO₂ has been calculated as a function of fission rate, irradiation dose, and temperature, taking into account the re-solution phenomenon. In these calculations, the experimentally determined re-solution

parameter of 10^{-4} sec⁻¹ for a fission rate of 10^{13} fissions/cm³ · sec is used, and the re-solution parameter is varied linearly with fission rate. Based on these assumptions, ≈ 90% of the fission gases are predicted to remain in solution at 1400°C up to 14 at % burnup under a fission rate of 10^{14} fissions/cm³ · sec. Thus, the gaseous fission-product contribution to fuel swelling can be calculated by assuming that essentially all the fission gases are present in the atomic state. The collection of fission gas at grain boundaries, which leads to interconnection of intergranular bubbles and eventual gas release, is expected to be controlled by either re-solution and atomic diffusion or diffusion of small clusters of fission-gas atoms, rather than by the migration of bubbles up the temperature gradient.

Since a large volume of the fuel is irradiated at temperatures below 1400°C under typical light-water reactor (LWR) and fast breeder reactor (FBR) operating conditions, we conclude that the re-solution process plays an important role in controlling gaseous fission-product swelling and gas-release behavior in both LWR and FBR fuel.

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